

Xiayang Jin

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EDUCATION

University of Michigan, Ann Arbor

Bachelor's of Science, Computer Science

- Cumulative GPA: 4.0/4.0

Aug 2024 — May 2026

Ann Arbor, MI

Shanghai Jiaotong University

Bachelor's of Science, Mechanical Engineering

Sep 2022 — Now

Shanghai, China

EXPERIENCES

Research Intern

OrderLab, University of Michigan

May 2025 — May 2026

Ann Arbor, MI

- Conduct research at the System-ML boundary on reliability and performance of machine learning systems
- Develop system support for failure modeling, tolerance and recovery strategies of ML workloads that preserve correctness while minimizing overhead
- Evaluate techniques on LLM/RLHF workloads across multi-GPU setups; analyze trade-offs among time-to-recovery, throughput, and data integrity

Research Intern

Shanghai Jiaotong University

May 2024 — Aug 2024

Shanghai, China

- Research in MINLP optimizations and their applications in Computer Networking
- Designed an Machine Learning approach to boost traditional MINLP solvers, and applied it to cutting-edge networking algorithms

Instructional Aide

University of Michigan

Jan 2026 — May 2026

Ann Arbor, MI

- Assisted in presenting a course about Introduction to Operating Systems
- Conducted weekly lab and office hours for over 200 students

Teaching Assistant

Shanghai Jiaotong University

May 2024 — Aug 2024

Shanghai, China

- Assisted in presenting a course about game development and software engineering
- Developed auto grading and feedback programs for student projects
- Conducted weekly lab and office hours for over 50 students

PROJECTS

PhoenixML

May 2025 — May 2026

- Developed a stage-level automatic recovery system for distributed machine learning training jobs with transient CUDA faults without checkpoint/restore
- Designed an runtime interposition layer to transparently trace CUDA alloc/launch calls and maintain a VA to tensor registry
- Implemented CUDA Virtual Memory Management with a proxy context to enable zero-copy remapping across context respawns

CudaProxy

Aug 2025 — May 2026

- Developed a CUDA runtime proxy for machine learning inference to use CUDA Graphs and Persistent Kernels for accelerated performance without brittle fusion
- Implemented Bucketing, Automatic Padding, and Static Memory Pooling to handle dynamic workloads with minimal overhead
- Implemented automatic routing for kernel launches based on workload characteristics

OS Kernel Implementation

Jan 2025 — May 2025

- Developed a C++ thread library and virtual memory pager on UNIX-like systems
- Implemented synchronization primitives like mutexes and condition variables using UNIX context management techniques.
- Managing Thread life cycle, Scheduler, Context Switching, CPU Booting, Process creation, Process forking, Page fault handling, and Process destruction
- Implemented optimizations including copy-on-write and page sharing

Dynamic Typed Compiler

May 2024 — Aug 2024

- Used Rust to develop a compiler for a simple language supporting dynamic typing and heap allocation on x86-64 architecture
- Implemented frontend checking, middleend Single Static Assignment (SSA) forming, and backend code generation with System V ABI
- Implemented optimizations including register allocation and assertion removal

Hypertile

May 2025 — Present

- Developed a high-performance tiling window management mechanism for Linux using C++ under Wayland protocol, targeting efficient multitasking across multiple projects
- Features include workspace management, windows overview, dynamic window tiling, and multi-monitor support

Deep Learning Approach to MINLP Problems in Wireless Networks

May 2024 — Aug 2024

- Designed and implemented an imitation learning approach to accelerate Branch and Bound algorithm for solving Mixed-Integer Nonlinear Programming optimization problems
- Applied the approach to Interference Graph Estimation in full-duplex millimeter-wave backhaul networks

SKILLS

- **Programming Languages:** C/C++, Python, CUDA, Rust, HTML, CSS, Java, SQL, Bash, Elm
- **Technologies:** Linux Kernel, CUDA, DeepSpeed, Pytorch, Tensorflow, Git, UNIX, Docker